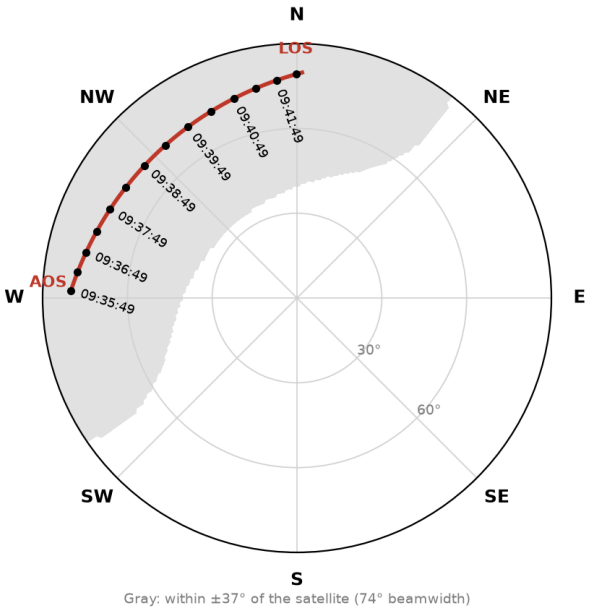
SO-50 Pass Plan

Monday 10/19/2026 Rise 09:35:49 Max 09:39:10 Set 09:42:32 Max El 19° Az 272°→2°

TX (uplink) 145.850 MHz FM, 67.0 Hz CTCSS **RX (downlink) 436.795 MHz FM** — Doppler-corrected, rounded to 5 kHz (error up to 2.5 kHz)

Time	TX MHz	RX MHz	Az	El
09:35:49	145.850	436.800	272°	10°
09:36:19	145.850	436.800	277°	12°
09:36:49	145.850	436.800	282°	14°
09:37:19	145.850	436.800	288°	15°
09:37:49	145.850	436.800	295°	17°
09:38:19	145.850	436.795	303°	18°
09:38:49	145.850	436.795	311°	19°
09:39:19	145.850	436.795	319°	19°
09:39:49	145.850	436.795	327°	18°
09:40:19	145.850	436.790	335°	17°
09:40:49	145.850	436.790	342°	16°
09:41:19	145.850	436.790	349°	14°
09:41:49	145.850	436.790	355°	13°
09:42:19	145.850	436.790	360°	11°



Gray: within ±37° of the satellite (74° beamwidth)

Yaesu VX-6R + speaker mic: uplink TX

- **Set Mode access:** [F/W] then [0(SET)], dial to the item, [0(SET)] to edit, dial = value, PTT = save
- **Frequency:** VHF band, VFO: key in digits for 145.850; the dial retunes for Doppler
- **Tone:** [F/W] then [MODE(SPS) SQ TYP], dial to TONE (not T SQL), PTT to save; [F/W] then [2(CODE)], dial to 67.0 Hz, press [2(CODE)] to save
- **Power:** [F/W] then [7(LK)TXPO] cycles HIGH / LOW3 (2.5 W) / LOW2 (1 W) / LOW1; use LOW3
- **Set Mode 1 APO:** OFF
- **Set Mode 4 ARS:** OFF
- **Set Mode 6 BCLO:** OFF, so PTT is never blocked by the downlink
- **Set Mode 27 HLF.DEV:** OFF: full deviation
- **Set Mode 37 MCGAIN:** LVL 5, adjust to your voice on the speaker mic
- **Set Mode 51 RPT:** SIMP (simplex); shortcut [F/W] [6(RPT)]
- **Set Mode 67 TOT:** 1 min (default 3)
- **Set Mode 70 TXSAVE:** OFF
- **Do not:** tap [7(LK)TXPO] alone: it turns on the Internet DTMF beep at every TX
- **Arming tone:** set tone to 74.4 Hz (as above) and key 2 s, then set 67.0 Hz and talk

Alinco DJ-VX50: downlink RX

- **Menu access:** FUNC, then 8, UP/DOWN to the menu number, dial = value, FUNC = confirm, ESC = exit
- **Frequency:** VFO mode: key in digits, e.g. 436.795 = 4 3 6 7 9 5; the dial then retunes for Doppler
- **Menu 01 R-CDC:** OFF (default): a tone squelch would mute the down-link
- **Menu 03 STEP:** 5K (default is 10K), so each dial click is one ladder step
- **Menu 04 W/N:** WIDE (default)
- **Menu 17 SAVE:** OFF (default is ON): battery saver can clip weak openings
- **Menu 19 SQL:** start at 1-2 (default 9 is tight); raise only if noise is constant
- **Menu 20 REPEAT:** OFF (default), and FUNC 4 until no offset is shown: simplex
- **Keypad lock:** FUNC, then hold #; the dial still retunes, and PTT stays untouched on this radio
- **Audio:** Earphone, volume up enough to hear a weak FM carrier

Notes

- ARM FIRST: 2 s carrier with 74.4 Hz tone, then switch to 67.0 Hz. The timer runs about 10 min, so arm just after rise (re-arm if it drops).
- Single-channel FM: short calls, no long conversations.

General

- **Identify:** KF8EEZ, short calls; low power. Keep both radios and the TX antenna well away from the SDR
- **Two radios:** Yaesu VX-6R + MH-57A4B speaker mic = uplink TX (log-periodic); Alinco DJ-VX50 = downlink RX. Use an earphone on the Alinco so its speaker doesn't feed the speaker mic
- **Aim (log-periodic):** point AT satellite: pre-aim at rise Az, follow the Az/El ladder; slant 45°